

Detecting COVID-19 Vaccine Misinformation on Twitter Using CT-BERT and Logistic Regression

Research Paper

Author	Nikhil Emmanuel Nanda
Project Type	Independent passion project
Focus	Binary detection of COVID-19 vaccine misinformation-supporting tweets

Abstract

The spread of COVID-19 vaccine misinformation on social media created major challenges for public understanding of vaccine safety, side effects, and conspiracy claims. [1] This project investigates whether natural language processing models can detect tweets that support misinformation versus tweets that do not support it. Using vaccine-related posts from the COVMis-Stance dataset, the original stance labels were simplified into a binary classification task: misinfo for tweets favoring misinformation and not_misinfo for tweets against or neutral toward misinformation. Two models were tested: a Logistic Regression baseline with TF-IDF features and a transformer-based model using COVID-Twitter-BERT (CT-BERT). On the held-out test set, Logistic Regression achieved 0.67 accuracy and 0.62 macro F1, while CT-BERT achieved 0.52 accuracy and 0.52 macro F1. Although CT-BERT performed strongly on the validation split, it did not generalize as well on the final test split. These findings show that more complex transformer models do not always outperform simpler baselines when the dataset is small. The project highlights the importance of dataset size, class balance, and careful evaluation in misinformation detection research.

Keywords: misinformation detection; COVID-19 vaccines; Twitter; CT-BERT; Logistic Regression; natural language processing

Study Overview

Component	Design Choice	Verified Value	Notes
Task framing	Binary classification	misinfo vs not_misinfo	The original stance task was simplified for the pilot study.
Source data	Vaccine-related tweets from COVMis-Stance	139 filtered tweets	After keyword filtering and label simplification.
Baseline	TF-IDF + Logistic Regression	0.67 accuracy, 0.62 macro F1	Best classical test result in this study.
Transformer	COVID-Twitter-BERT	0.52 accuracy, 0.52 macro F1	Strong validation result but weaker test generalization.

1. Introduction

During the COVID-19 pandemic, social media became one of the main places where vaccine information and misinformation spread rapidly. Many users shared claims about vaccine safety, ingredients, side effects, government surveillance, and other conspiracy theories. Because these posts can influence public opinion before experts or fact-checkers respond, it is important to study ways of identifying misinformation-related content early.

This project focuses on Twitter posts related to COVID-19 vaccines. Rather than trying to determine whether each tweet is objectively true or false, this study examines whether a tweet appears to support misinformation or not support misinformation. This approach is useful because it focuses on the language and stance expressed in the tweet itself.

The goal of this passion project was to compare a traditional machine learning baseline and a domain-specific transformer model for this task. Specifically, the study compares Logistic Regression with TF-IDF

features and CT-BERT, a transformer pretrained on COVID-related Twitter language. By comparing both models, the project evaluates whether the more advanced neural approach actually improves performance on a small pilot dataset.

2. Background

Detecting misinformation online is a difficult natural language processing problem. Many posts contain emotional language, sarcasm, repeated rumors, and short incomplete sentences. In health-related topics, misinformation can be especially harmful because readers may use it to make decisions about treatment or prevention.

A common approach to misinformation detection is text classification. Traditional machine learning models such as Logistic Regression often use TF-IDF features, which represent how important a word or phrase is in a document compared with the rest of the dataset. [3] These models are fast and interpretable, but they often struggle with context, sarcasm, and negation.

Transformer models such as BERT are more powerful because they understand words in context. CT-BERT, or COVID-Twitter-BERT, is especially relevant because it was pretrained on Twitter posts related to COVID-19. This makes it a good candidate for classifying vaccine-related tweets, since the model has already seen the style and vocabulary of pandemic-era Twitter language.

Even so, advanced models do not always perform better in every situation. When the dataset is small, a simpler baseline may sometimes generalize more reliably. This project tests that idea directly.

3. Dataset and Scope

This project used vaccine-related tweets from the COVMis-Stance dataset. [3] Tweets were filtered using vaccine-related keywords such as vaccine, pfizer, moderna, booster, jab, shot, and astrazeneca. Only English tweets relevant to COVID-19 vaccine misinformation were included.

The original labels were simplified into two classes. The misinfo class included tweets labeled favor, while the not_misinfo class combined tweets labeled against and neither. This means the final experiment was a binary classification task rather than a three-class stance classification task.

The tweet text was cleaned by removing URLs, usernames, hashtag symbols, and extra spaces, then converting all text to lowercase. After filtering and preprocessing, the dataset contained 139 vaccine-related tweets: 59 misinfo and 80 not_misinfo.

For the CT-BERT experiment, the data was split into 97 training samples, 21 validation samples, and 21 test samples. For the Logistic Regression baseline, a balanced pilot dataset of 118 tweets was used, with 59 tweets in each class.

4. Methodology

The first model tested was Logistic Regression using TF-IDF features. The vectorizer used up to 3000 features and included unigrams and bigrams. The classifier used a maximum of 200 iterations with balanced class weights. A grid search explored multiple values of C, solver type, and maximum iterations. The best hyperparameters were $C = 0.1$, solver = liblinear, and max_iter = 200, although the tuned model produced the same final test performance as the original baseline.

The second model used was `digitalepidemiologylab/covid-twitter-bert-v2`, referred to as CT-BERT. [4] The model was loaded for binary sequence classification with two output labels. Tweets were tokenized to a maximum length of 128 tokens, and dynamic padding was applied through a data collator.

The CT-BERT training configuration used a learning rate of $2e-5$, a batch size of 8, weight decay of 0.01, and early stopping with patience set to 2. The best model checkpoint was selected using macro F1. Evaluation focused on accuracy, macro precision, macro recall, macro F1, and confusion matrices.

5. Results

The baseline and transformer models produced different performance patterns. Logistic Regression delivered the strongest held-out test result, while CT-BERT looked promising on validation but dropped on the final test split.

Model	Split	Accuracy	Macro Precision	Macro Recall	Macro F1
Logistic Regression	Test	0.67	0.80	0.67	0.62
CT-BERT	Validation	0.8571	0.8606	0.8472	0.8518
CT-BERT	Test	0.52	0.54	0.54	0.52

5.1 Logistic Regression test performance

On the balanced pilot test set, the Logistic Regression baseline achieved 0.67 accuracy and 0.62 macro F1. Class-wise performance showed an F1 of 0.75 for the `misinfo` class and 0.50 for the `not_misinfo` class. The confusion matrix $\begin{bmatrix} 15 & 0 \\ 10 & 5 \end{bmatrix}$ shows that the model correctly identified all 15 misinformation-supporting tweets in the test set, but it misclassified 10 of the 15 `not_misinfo` tweets as misinformation-supporting.

5.2 CT-BERT validation and test performance

On the validation set, CT-BERT achieved 0.8571 accuracy and 0.8518 macro F1, suggesting that the model learned useful patterns during fine-tuning. However, test performance was lower: 0.52 accuracy and 0.52 macro F1. Class-wise test performance was 0.55 F1 for the `misinfo` class and 0.50 F1 for the `not_misinfo` class. The confusion matrix $\begin{bmatrix} 6 & 3 \\ 7 & 5 \end{bmatrix}$ shows that CT-BERT correctly classified 6 of 9 `misinfo` tweets and 5 of 12 `not_misinfo` tweets.

5.3 Direct comparison

In this pilot study, Logistic Regression outperformed CT-BERT on the held-out test set. This result does not mean transformer models are inherently worse for misinformation detection. Instead, it suggests that a small dataset can prevent a higher-capacity model from showing its full advantage.

6. Discussion

The results were informative because they did not fully match the original expectation. Since CT-BERT is a transformer model pretrained on COVID-related Twitter language, it seemed likely that it would outperform a simple Logistic Regression model. However, in this project, the Logistic Regression baseline achieved better test performance.

One major reason is likely the small dataset size. CT-BERT was trained on only 97 training tweets, with 21 tweets in the validation set and 21 tweets in the test set. With such a small dataset, even a few incorrect

predictions can significantly change the final scores. Transformers usually benefit from larger datasets because they contain many parameters and can overfit when training data is limited.

A second issue is that tweets supporting misinformation and tweets refuting misinformation often use the same vocabulary. A supportive tweet and a rebuttal tweet may both mention words like vaccine, autism, microchip, or Bill Gates. The difference depends on context, negation, and tone. Even though CT-BERT is better at context than TF-IDF, the small sample size may have prevented it from learning these distinctions reliably.

Another likely factor is overfitting. CT-BERT achieved very strong validation results but then dropped to 0.52 macro F1 on the final test set. This gap suggests that the model learned patterns specific to the validation split rather than patterns that generalized well to new examples.

7. Error Analysis

Three main error patterns emerged from the confusion matrices and class-wise scores.

First, negation and rebuttal language created difficulty for both models. Tweets that reject misinformation often contain the same misinformation-related words as tweets that promote it, which can blur the decision boundary.

Second, shared topic vocabulary caused class overlap. Both classes discussed vaccine-related rumors, so the models sometimes learned topic words more strongly than stance cues.

Third, the limited amount of training data made the task unstable. The dataset was too small to represent the wide range of tweet styles, including sarcasm, short fragments, uncertainty, and emotionally charged language.

Future improvements could include collecting more labeled data, using cross-validation instead of one small split, preserving the original three-class stance structure, and exploring stronger augmentation or class-weighted training.

8. Limitations and Ethical Considerations

This project has several limitations. The dataset was very small, which reduces reliability and makes the final results sensitive to the exact train-validation-test split. The experiment also simplified the task into two classes, which loses nuance between neutral tweets and tweets that actively refute misinformation. In addition, the tweets came from a specific COVID-era context and may not generalize to newer health misinformation topics or other platforms.

This paper is intended for educational research only. Even if a model can identify misinformation-supporting language, it should not be used as a fully automatic moderation tool without human review. Health communication is a sensitive area, and automated systems can make mistakes when sarcasm, ambiguity, or context are involved.

To reduce privacy concerns, preprocessing removed usernames and URLs, and the analysis focused on textual patterns rather than identifying individuals. A model prediction in this setting should be interpreted as a classification of language patterns, not as final proof of medical truth.

9. Future Work

Several next steps would strengthen this project. The most important improvement is expanding the dataset with more labeled tweets. A larger corpus would likely help CT-BERT generalize more reliably.

A second improvement is restoring the original three-class stance structure: supports misinformation, refutes misinformation, and neutral or unclear. That design would better match the original research idea and produce a more nuanced analysis.

Additional experiments could compare multiple transformer models, use cross-validation, apply class-weighted losses, and test whether emerging misinformation themes can be identified before formal fact-checking.

10. Conclusion

This passion project explored the detection of COVID-19 vaccine misinformation-related tweets using both a traditional machine learning baseline and a transformer-based model. Tweets from the COVMis-Stance dataset were filtered, cleaned, and simplified into a binary classification task: misinfo versus not_misinfo.

The Logistic Regression baseline achieved 0.67 accuracy and 0.62 macro F1, while CT-BERT achieved 0.52 accuracy and 0.52 macro F1 on the final test set. Although CT-BERT showed strong validation performance, it did not generalize as well as expected. In this pilot experiment, the simpler Logistic Regression model performed better on held-out test data.

These results show that advanced models are not always better when the dataset is small. The project demonstrates both the promise and the difficulty of misinformation detection on social media. With more data, better balancing, and a richer stance-based setup, this work could be extended into a stronger and more realistic misinformation detection system.

Appendix A. Core Experimental Values

Setting	Value	Comment
Tokenizer length	128 tokens	Used for CT-BERT tweet inputs
CT-BERT batch size	8	Fine-tuning setting
CT-BERT learning rate	2e-5	Fine-tuning setting
CT-BERT early stopping	Patience = 2	Validation-based
Logistic Regression features	TF-IDF unigrams + bigrams	Max features = 3000
Pilot baseline set	118 tweets	Balanced: 59 per class

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